

Qiao Gu

PH.D. STUDENT · UNIVERSITY OF TORONTO

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Education

University of Toronto

PH.D. IN COMPUTER SCIENCE

Toronto ON, Canada

09/2021 - 09/2026 (expected)

- GPA: 4.0 / 4.0
- Advisor: [Prof. Florian Shkurti](#)
- Research Interest: 3D Computer Vision, Robot Learning, Generative Models

Carnegie Mellon University

M.S. IN ROBOTICS

Pittsburgh PA, U.S.A.

08/2019 - 08/2021

- GPA: 4.24 / 4.33
- Advisor: [Prof. David Held](#) and [Prof. Martial Hebert](#)

Hong Kong University of Science and Technology

B.ENG. IN COMPUTER SCIENCE AND ELECTRONIC ENGINEERING

Hong Kong S.A.R.

09/2015 - 06/2019

- GPA: 4.15 / 4.30
- Advisors: [Prof. Chi-Keung Tang](#) and [Prof. Yu-Wing Tai](#)

Professional Experience

Mar 2026 | **Research Scientist Intern | NVIDIA, TORONTO ON., Canada**
Aug 2026 | With [Jiahui Huang](#) and [Sanja Fidler](#)

Video Generation | World Modeling | 3D Reconstruction

June 2025 | **Research Scientist Intern | Meta Reality Labs, REDMOND WA, U.S.A.**
Feb 2026 | With [Julian Straub](#)

Video Generation with Spatial Memory | Egocentric World Models

Aug 2023 | **Research Scientist Intern | Meta Reality Labs, REDMOND WA, U.S.A.**
Feb 2024 | With [Chris Sweeney](#) and [Zhaoyang Lv](#)

Open-world Segmentation | 3D Gaussian Splatting | Egocentric Perception

Selected Projects

E³C: Video generation with 3D spatial memory and ego-exo pose control *Preprint, First Author (Meta Internship)*

- Proposed a controllable **latent video diffusion** model that renders an explicit **SLAM-based 3D environmental memory** to maintain viewpoint-consistent egocentric generation.
- Augmented the 3D memory with per-point **VAE appearance features** and fused them via a context adapter to reduce texture drift and improve object consistency.
- Implemented **ego-exo pose control** (exo 2D skeletons; ego 3D joints + 6DoF wrists) with **persistent pose tokens** for occlusion-robust motion adherence and controllable scene edits.
- **Set best-reported Nymeria results** with **249 FVD** (-24% vs best fine-tuned baseline) and **2.4 cm** camera translation error.

SAFE: Scalable failure estimation for vision-language-action models

NeurIPS 2025, First Author

- Formulated **multitask failure detection** for VLA policies by learning a task-agnostic failure signal directly from internal model representations.
- Trained lightweight temporal predictors (MLP/LSTM) to output calibrated **failure likelihood** scores with minimal runtime overhead and no policy modifications.

- Applied **functional conformal prediction** to calibrate time-varying thresholds, enabling reliable intervention trade-offs across tasks and embodiments.
- Reached **88.4 ROC-AUC** on *unseen* real-robot tasks (OpenVLA/WidowX), improving +18.4 over the strongest baseline.

EgoLifter: Open-world 3D segmentation for egocentric perception

ECCV 2024, First Author (Meta Internship)

- Built an **open-world 3D instance segmentation** system on top of **3D Gaussian Splatting**, learning per-Gaussian embeddings for object-level decomposition.
- Introduced a self-supervised **transient prediction** module to filter dynamic regions during reconstruction, reducing floaters and sharpening instance features.
- Lifted **SAM** masks into 3D via contrastive learning, enabling promptable 2D/3D segmentation and object-level extraction for scene editing.
- Introduced EgoLifter and an ADT benchmark; achieved **48.8** cross-view **mIoU** (+27.3 vs Gaussian Grouping) for open-world 3D segmentation from egocentric video.

ConceptGraphs: Open-vocabulary 3D scene graphs for perception & planning

ICRA 2024, Co-first Author

- Implemented an **object-centric 3D mapping** pipeline that constructs **open-vocabulary scene graphs** for downstream perception and planning.
- Engineered robust **localization** and **map update** mechanisms (re-association, persistence, change handling) to maintain consistent object maps over time.
- Prototyped **object captioning** and ran targeted **segmentation/association** studies to improve object proposals and multi-view consistency.

Publications

** Indicates equal contribution.*

E³C: Video Generation with 3D Environmental Memory and Ego-Exo Human Pose Control

Preprint

Qiao Gu, Adam Harley, Lingni Ma, Florian Shkurti, Julian Straub

SAFE: Scalable Failure Estimation for Vision-Language-Action Models

NeurIPS 2025

Qiao Gu, Yuanliang Ju, Shengxiang Sun, Igor Gilitschenski, Haruki Nishimura, Masha Itkina, Florian Shkurti

[Paper](#) [Website](#) [Code](#)

MomaGraph: State-Aware Unified Scene Graphs with Vision-Language Models for Embodied Task Planning

ICLR 2026 (Oral)

Yuanchen Ju, Yongyuan Liang, Yen-Jen Wang, Gireesh Nandiraju, Yuanliang Ju, Seungjae Lee, **Qiao Gu**, Elvis Hsieh, Furong Huang, Koushil Sreenath

[Paper](#) [Website](#)

SurgSemGS: Semantic and Contrastive Gaussian Splatting for Dynamic Surgical Scene Understanding

Preprint

Liyuan Liu, **Qiao Gu**, Daniel Elson, Baoru Huang

SICNav-Diffusion: Safe and Interactive Crowd Navigation with Diffusion Trajectory Predictions

RA-L & ICRA 2026

Sepehr Samavi, Anthony Lem, Fumiaki Sato, Sirui Chen, **Qiao Gu**, Keijiro Yano, Angela P Schoellig, Florian Shkurti

[Paper](#)

EgoLifter: Open-world 3D Segmentation for Egocentric Perception

ECCV 2024

Qiao Gu, Zhaoyang Lv, Duncan Frost, Simon Green, Julian Straub, Chris Sweeney

[Paper](#) [Website](#) [Code](#)

Aria Everyday Activities Dataset

arXiv 2024

Zhaoyang Lv, Nicholas Charron, Pierre Moulon, Alexander Gamino, Cheng Peng, Chris Sweeney, Edward Miller, Huixuan Tang, Jeff Meissner, Jing Dong, Kiran Somasundaram, Luis Pesqueira, Mark Schwesinger, Omkar Parkhi, **Qiao Gu**, Renzo De Nardi, Shangyi Cheng, Steve Saarinen, Vijay Baiyya, Yuyang Zou, Richard Newcombe, Jakob Julian Engel, Xiaqing Pan, Carl Ren

[Paper](#) [Website](#)

ConceptGraphs: Open-vocabulary 3D Scenegraps for Perception and Planning

ICRA 2024

Qiao Gu^{*}, Alihusein Kuwajerwala^{*}, Sacha Morin^{*}, Krishna Murthy Jatavallabhula^{*}, Bipasha Sen, Aditya Agarwal, Corban Rivera, William Paul, Kirsty Ellis, Rama Chellappa, Chuang Gan, Celso Miguel de Melo, Joshua B. Tenenbaum, Antonio Torralba, Florian Shkurti, Liam Paull

[Paper](#) [Website](#) [Code](#)

ConceptFusion: Open-set Multimodal 3D Mapping

RSS 2023

Krishna Murthy Jatavallabhula, Alihusein Kuwajerwala^{*}, **Qiao Gu**^{*}, Mohd Omama^{*}, Tao Chen, Shuang Li, Ganesh Iyer, Soroush Saryazdi, Nikhil Keetha, Ayush Tewari, Joshua B. Tenenbaum, Celso Miguel de Melo, Madhava Krishna, Liam Paull, Florian Shkurti, Antonio Torralba

[Paper](#) [Website](#) [Code](#)

Preserving Linear Separability in Continual Learning by Backward Feature Projection

CVPR 2023

Qiao Gu, Dongsub Shim, Florian Shkurti

[Paper](#) [Code](#)

OSSID: Online Self-supervised Instance Detection by (and for) Pose Estimation

RA-L & ICRA 2022

Qiao Gu, Brian Okorn, David Held

[Paper](#) [Website](#) [Code](#)

ZePhyR: Zero-shot Pose Hypothesis Rating

ICRA 2021

Brian Okorn^{*}, **Qiao Gu**^{*}, Martial Hebert, David Held

[Paper](#) [Website](#) [Code](#)

Deep Video Matting via Spatio-Temporal Alignment and Aggregation

CVPR 2021

Yanan Sun, Guanzhi Wang^{*}, **Qiao Gu**^{*}, Chi-Keung Tang, Yu-Wing Tai

[Paper](#)

iQUANT: Interactive Quantitative Investment Using Sparse Regression Factors

EuroVis 2021

Xuanwu Yue, **Qiao Gu**, Deyun Wang, Huamin Qu, Yong Wang

[Paper](#)

LADN: Local Adversarial Disentangling Network for Facial Makeup and De-Makeup

ICCV 2019

Qiao Gu^{*}, Guanzhi Wang^{*}, Mang Tik Chiu, Yu-Wing Tai, Chi-Keung Tang

[Paper](#) [Website](#) [Code](#)

Characterizing Fluid Response and Sepsis Progression in Emergency Department Patients

EMBC 2019

Qiao Gu, Varesh Prasad, Thomas Heldt

[Paper](#)

PreserVis, a Visual Analytic System for Traffic and Pollution Patterns

VAST 2017

Qiao Gu, Hang Yin, Lian Chen, Haotian Li, Chengzhong Liu, Xuanwu Yue, Huamin Qu

[Paper](#)

Awards, Scholarships, & Grants

2025 Mary H. Beatty Fellowship

CAD 10,000

2025 University Nomination for Google PhD Fellowship

2025 Finalist of Qualcomm Innovation Fellowship

2025	Kwok Sau Po Scholarship	CAD 1,750
2025	The Fahiem Bacchus Memorial Graduate Student Scholarship	CAD 2,000
2024	NeurIPS 2024 Top Reviewer	
2024	Mary H. Beatty Fellowship	CAD 10,000
2024	Ray Reiter Graduate Award	CAD 1,000
2023	Acres Productive Technologies Inc./Joseph Yonan Memorial Fellowship	CAD 2,000
2022	Ontario Graduate Scholarships	CAD 15,000
2019	HKUST Academic Achievement Medal (Top 1%)	
2017-2019	HKSAR Government Scholarships	HKD 160,000
2015-2019	Dean's List	
2018	Mr. Armin and Mrs. Lillian Kitchell Undergraduate Research Award	
2017	High Fashion Charitable Foundation Exchange Scholarships	HKD 50,000
2017	HKSAR Government Scholarship Fund - Reaching Out Award	HKD 10,000
2017	HKSAR Government Scholarship Fund - Talent Development Scholarship	HKD 10,000
2016	Simatelex Charitable Foundation Scholarship	HKD 20,000
2016	University's Scholarship Scheme for Continuing Undergraduate Students	HKD 20,000
2015	University Admission Scholarship	HKD 40,000

Invited Talks

03/2024. *ConceptGraphs: Open-Vocabulary 3D Scene Graphs for Perception and Planning*. Boston Dynamic AI institute

02/2024. *Open-World 3d Segmentation with Project Aria*. Meta Reality Labs

10/2023. *ConceptGraphs: Open-Vocabulary 3D Scene Graphs for Perception and Planning*. Meta Reality Labs

Teaching Experience

Fall 2022	CSC384: Intro to Artificial Intelligence	Teaching Assistant
Winter 2022	CSC384: Intro to Artificial Intelligence	Teaching Assistant
Fall 2021	CSC110: Foundations of Computer Science I	Teaching Assistant

Mentorship

2025-present	Zhengyang Liang	PhD student, University of Toronto
2024-present	James Ross	Master student, University of Toronto
2024-present	Yuanliang Ju	PhD student, University of Toronto
2024-2025	Owen Sun	Undergraduate student, University of Toronto
2022-2025	Anthony Lem	Master student, University of Toronto
2023-2024	Sirui Chen	Undergraduate student, University of Toronto
2023	Edward Li	Undergraduate student, University of Toronto
2023	Spencer Li	Undergraduate student, University of Toronto

Outreach & Professional Development

SERVICE AND OUTREACH

2024-2025	UofT RI Reading Group	Organizer
2022	Toronto AI in Robotics Seminar	Organizer

REVIEWER SERVICE

TRO, TPAMI, IJRR, TMLR, RA-L, NeurIPS, R:SS, CVPR, ICLR, ICCV, ICRA, IROS, ECCV, ICML, WACV
PROFESSIONAL MEMBERSHIPS

Student member, IEEE